Problem 3.0.23

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Problem

Let $x_1 = 3.5$, $x_2 = 7.5$ and $x_3 = 5.2$ be observed values of a random sample of size three from a population having uniform distribution over the interval $(\theta, \theta + 5)$, where $\theta \in (0, \infty)$ is unknown and is to be estimated. Then which of the following is NOT a maximum likelihood estimate of θ?

1. 2.4
2. 2.7
3. 3
4. 3.3

Representation

For a random sample $x_1, x_2, \ldots, x_n$ from a uniform distribution $U(\theta, \theta + 5)$, the probability density function (PDF) for each observation is:

$$f(x_i|\theta) = \begin{cases} \frac{1}{5}, & \theta < x_i < \theta + 5 \\ 0, & \text{otherwise} \end{cases} \tag{1.1}$$

The likelihood function $L(\theta)$ for the sample of size $n = 3$ is the product of the individual PDFs:

$$L(\theta) = \prod_{i=1}^{3} f(x_i|\theta) = \begin{cases} \left(\frac{1}{5}\right)^3 = \frac{1}{125} & \text{if } \theta \leq x_i \leq \theta + 5 \text{ for all } i \\ 0 & \text{otherwise} \end{cases} \tag{1.2}$$

Constraints on θ

θ must be less than or equal to all observed values:

$$\theta \leq \min(x_1, x_2, x_3) \tag{1.3}$$
$$\theta \leq 3.5 \tag{1.4}$$

Also the upper limit must be greater than or equal to all observed values, Hence

$$\theta + 5 \geq \max(x_1, x_2, x_3) \tag{1.5}$$
$$\theta + 5 \geq 7.5 \tag{1.6}$$
$$\theta \geq 7.5 - 5 \tag{1.7}$$
$$\theta \geq 2.5 \tag{1.8}$$

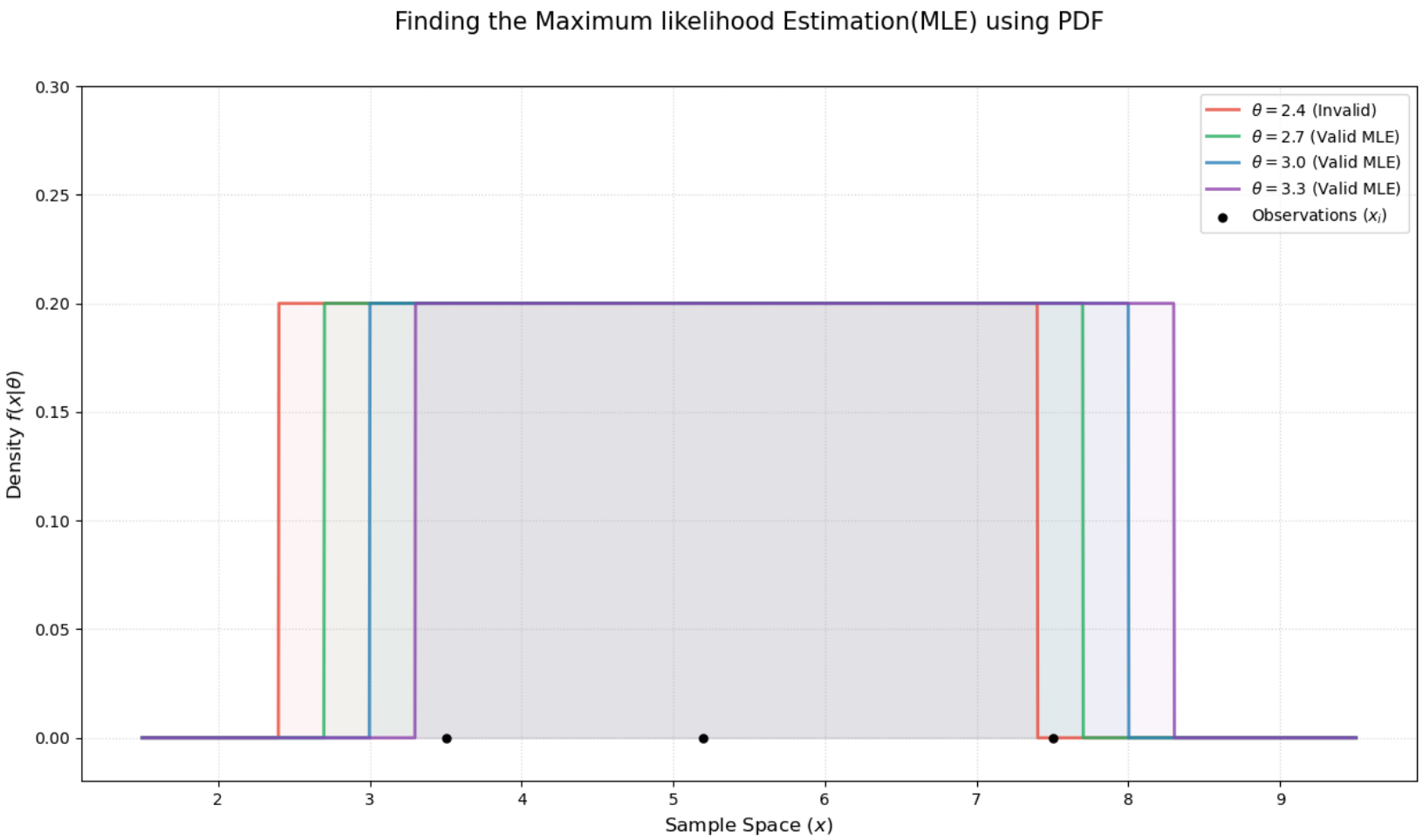
Calculation

Combining both the constraints, We get the range of θ

$$2.5 \leq \theta \leq 3.5 \tag{1.9}$$

From the options, 2.7, 3, 3.3 lies in range of θ but 2.4 doesn't.
Hence 2.4 is NOT a maximum likelihood estimate of θ.

Plot


Finding the Maximum likelihood Estimation(MLE) using PDF
$\theta = 2.4$ (Invalid)
$\theta = 2.7$ (Valid MLE)
$\theta = 3.0$ (Valid MLE)
$\theta = 3.3$ (Valid MLE)
Observations (x_i)
Density $f(x|\theta)$
Sample Space (x)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
2
3
4
5
6
7
8
9

Python Code for Plotting

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import uniform

# Setup figure
fig, ax = plt.subplots(figsize=(14, 8))

# Define our observed data
data = np.array([3.5, 5.2, 7.5])

# Define the candidate thetas to test and their colors
thetas =[2.4, 2.7, 3.0, 3.3]
colors = ['#e74c3c', '#27ae60', '#2980b9', '#8e44ad']

x = np.linspace(1.2, 9.8, 5000)

# Loop through, calculate validity, and plot using SciPy
for theta, color in zip(thetas, colors):
```

Python Code for Plotting

```
    is_valid = np.all((data >= theta) & (data <= theta + 5))

    status = Valid MLE if is_valid else Invalid
    dynamic_label = rf'$\theta = {theta}$ ({status})'

    # Calculate PDF using SciPy
    y = uniform.pdf(x, loc=theta, scale=5)

    # Plot the outline calculated by SciPy
    ax.plot(x, y, color=color, linewidth=2.2, alpha=0.9, label=
        dynamic_label)

    ax.fill_between(x, y, 0, color=color, alpha=0.08)

# Plot the observed data points
ax.scatter(data, [0, 0, 0], color='black', s=35, zorder=10,
           label=r'Observations ($x_i$)')
```

Python Code for Plotting

```
# Formatting
ax.set_title('Finding the Maximum likelihood Estimation (MLE)
    using PDF',
             fontsize=16, pad=25)
ax.set_xlabel(r'Sample Space ($x$)', fontsize=12)
ax.set_ylabel(r'Density $f(x|\theta)$', fontsize=12)

# Set axis limits
ax.set_ylim(-0.02, 0.30)

# Add a light dotted grid
ax.grid(True, linestyle=':', alpha=0.5)

# Add the legend
ax.legend(loc='upper right', framealpha=1.0, fontsize=10.5)

plt.tight_layout()
plt.show()
```